

Chinese yam polysaccharide enhances growth performance and cellular immune response in weanling rats

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Abstract

BACKGROUND: In order to develop a novel dietary additive for potential application in domestic animals and humans, a fraction of Chinese yam polysaccharide (CYP) was prepared and a study was conducted to determine the effects of dietary supplementation with CYP on the immune response and growth performance in weanling rats.

RESULTS: Dietary supplementation with CYP increased the average daily feed intake and average daily gain but decreased the feed/gain ratio overall compared with the non-supplemented group. CYP also significantly increased the serum contents of interleukin-2 and tumour necrosis factor and the proliferation activity of peripheral lymphocytes but decreased the serum contents of low-density lipoprotein, total cholesterol and triglyceride compared with the non-supplemented group.

CONCLUSION: The increase in cellular immunity in CYP-supplemented weanling rats confers an important protective role in the non-specific defence against infections. This herbal polysaccharide, as a natural green dietary additive for promoting the immune response and healthy growth, may offer an effective alternative to antibiotics for weanling animals.

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Keywords: Chinese yam polysaccharide; immune response; performance

INTRODUCTION

It is well known that the abrupt changes in feed composition and feeding conditions at weaning cause a dramatic change in digestive function, which often results in the intestinal malabsorption of nutrients.¹ Because of the above changes, early weaning interrupts the supply of immunologically important factors (e.g. glutamine and arginine) from maternal milk.² Reduced feed intake and efficiency³ impair the production of antibodies and compromise cellular immune functions.⁴ Thus early weaning increases the susceptibility of young animals to Gram-negative bacterial infection (e.g. *Escherichia coli* infection) and disease incidence (e.g. diarrhoea). Therefore weaning is associated with growth retardation as well as an increase in both morbidity and mortality in neonatal animals.⁵

Weaning with widespread diarrhoea can lead to considerable economic loss, which can be counteracted by nutritional manipulation.⁶ Antibiotics have been traditionally used to prevent and treat enteric diseases (including diarrhoea) induced by weaning stress. However, the continuous use and misuse of antibiotics have led to the emergence of drug and antibiotic residues in animal products.⁷ Therefore novel feed additives are needed to replace antibiotics. As potential alternatives to antimicrobial agents, many Chinese herbal ingredients have been reported to have immunity-enhancing effects in humans and animals.^{8–12}

Growing evidence shows that phytochemicals have great potential for use in animal production.^{13–15} Kong *et al.*^{11,13} reported that Chinese herbal ultrafine powder and *Acanthopanax senticosus* extract as dietary additives enhance the cellular and humoral immune responses of weaned piglets by modulating the production of immunocytes, cytokines (including interleukin-1 β (IL-1 β), IL-2 and IL-6) and antibodies (including immunoglobulin G (IgG) and IgM). Chinese yam, the rhizome of various species of the genus *Dioscorea opposita* Thunb. (Dioscoreaceae), has been used as an important invigorant in traditional Chinese medicine

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for many years to strengthen stomach function, alleviate anorexia and cure diarrhoea.¹⁶ Starch, as one of the major polysaccharides, is the most abundant carbohydrate (200–600 g kg⁻¹) in this kind of herb.¹⁷ The granule size of native Chinese yam starch is variable, ranging from 5 to 50 µm. The granule surface appears smooth, with no evidence of any fissures or pores, and is disc-like or oval in shape, and a fairly large number of large disc-like granules are present. Native Chinese yam starch is a typical C-type starch, i.e. a mixture of A-type and B-type starches.^{17,18} However, there have been few investigations on the physiological properties of starch present in medicinal plants, especially Chinese yam. Research on the properties of starch in Chinese yam is very important owing to its ready availability and its potentially extensive utilisation in the food and medicine industries.

In order to develop novel immunomodulatory dietary additives for potential application in domestic animals and humans, four fractions of Chinese yam polysaccharide (CYP 1, CYP 2, CYP 3 and CYP 4) were characterised *in vitro* and *in vivo* in a previous study. Our findings suggested that CYP 3, as a phytochemical dietary additive, is a good source of carbon and energy and may improve bacterial community diversity and modulate short-chain fatty acid production in the hindgut of rats (Kong X-F *et al.*, unpublished). On the basis of the foregoing, we hypothesised that CYP 3 as a dietary additive may enhance the immune response and health and thus improve the growth performance in weanling rats. This hypothesis was tested in the present study by analysing the serum contents of biochemical metabolites and cytokines, the proliferation activity of peripheral blood lymphocytes as well as the growth performance on days 7 and 21 after initiation of dietary supplementation with CYP 3 in weanling rats.

MATERIALS AND METHODS

Preparation of CYP 3

The dried and sliced rhizome of Chinese yam (*Rhizoma dioscoreae*, *D. opposita*, from Henan Province, China) was decocted twice with distilled water (200 g L⁻¹) for 1 h each time. The two decoctions were pooled and condensed under hypopresia conditions to 1 g herb mL⁻¹ decoction. The condensed solution was purified as follows: (1) the solution was first deproteinised by adding 100 g L⁻¹ trichloroacetic acid dropwise until pH 3; (2) after overnight aging, the supernatant fluid isolated by centrifugation at 3000 × g for 10 min at 4 °C was extracted by precipitation with 450 mL L⁻¹ dehydrated ethanol;⁸ (3) the supernatant fluid was collected by centrifugation and precipitated with dehydrated ethanol at a final content of 600 mL L⁻¹ in the supernatant fluid; (4) the precipitates isolated by centrifugation were redissolved in distilled water (200 g L⁻¹) and re-precipitated with 600 mL L⁻¹ dehydrated ethanol. The polysaccharide (CYP 3) was obtained after lyophilising the precipitate collected when the final content of ethanol was 600 mL L⁻¹. The carbohydrate content of CYP 3 was measured as 909.1 g kg⁻¹ by the vitriol–anthrone method with anhydrous glucose as standard control.⁸ Protein and nuclear acid contamination in the extract was negligible (absorbances at 280 and 260 nm close to zero). The final extract was used as feed additive in this study, after the amount of endotoxin (less than 0.5 EU mL⁻¹) was up to the Chinese Veterinary Pharmacopoeia standard.¹⁹

Animals, housing and experimental design

Forty weanling Sprague-Dawley rats with an average initial body weight of 87 ± 3 g were randomly assigned to four treatment

Table 1. Dietary ingredients and nutrient levels of basal diet (g kg⁻¹, as fed basis)

Component	Amount
<i>Dietary ingredients</i>	
Maize (87 g crude protein (CP) kg ⁻¹)	380
Soybean meal (430 g CP kg ⁻¹)	370
Wheat bran (157 g CP kg ⁻¹)	130
Fish meal (625 g CP kg ⁻¹)	50
Soybean oil	30
NaCl	10
Mineral mixture ^a	15
Vitamin mixture ^b	15
<i>Nutrient levels^c</i>	
Digestible energy (MJ kg ⁻¹)	14.2
Crude protein	243
Ether extract	65
Crude fibre	36
Crude ash	76
Total Ca	11.6
Total P	8.0

^a Provided the following (kg⁻¹ diet): Cu, 13 mg; Fe, 270 mg; Mn, 64 mg; Zn, 70 mg; I, 0.8 mg; Se, 0.27 mg; Co, 0.6 mg.

^b Provided the following (kg⁻¹ diet): vitamin A, 14 000 IU; vitamin D₃, 1500 IU; vitamin E, 120 IU; vitamin K₃, 5 mg; thiamine, 13 mg; riboflavin, 12 mg; pyridoxine, 12 mg; vitamin B₁₂, 0.022 mg; biotin, 0.2 mg; D-pantothenic acid, 24 mg; folic acid, 6 mg; niacin, 60 mg; choline, 1250 mg.

^c Analysed values for all dietary components.

groups in a randomised complete block design representing 0, 0.25, 0.5 and 1 g kg⁻¹ CYP 3 supplementation to a maize- and soybean meal-based diet (Table 1). The basal diet was formulated to meet or exceed the nutrient requirements of growing rats according to Laboratory Animals – Mice and Rats Formula Feeds (GB 14 924.3-2001, China). There were ten rats (five males and five females) per treatment group, with one rat per cage. Each 50 cm × 36 cm × 28 cm cage was equipped with a poly-hole feeder and a water nipple to allow *ad libitum* consumption of feed and water. The temperature and relative humidity were maintained at 28 ± 2 °C and 65–75% respectively in an air-controlled nursery room.

The study was carried out in accordance with the Chinese guidelines for animal welfare and experimental protocol approved by the Animal Care Committee, Institute of Subtropical Agriculture, The Chinese Academy of Sciences, Changsha, China.²⁰

Growth performance

Body weights and feed intakes were measured at the beginning of the trial and weekly thereafter. On the basis of these data, average daily feed intake (g day⁻¹), average daily gain (g day⁻¹) and feed/gain ratio were calculated for the periods spanning days 0–7 and days 0–21.

Blood sampling and preparation

On days 7 and 21 after consumption of the experimental diets, four rats (two males and two females) were sampled randomly from each treatment group. Blood samples (5 mL per rat) were collected by removing the eyeball under general anaesthesia (ethyl ether inhalation) between 08:00 and 09:00 before feeding.

A 2 mL aliquot of each sample was immediately transferred into an aseptic capped tube containing sodium heparin for analysis of the effects of CYP 3 on lymphocyte proliferation activity. The remaining 3 mL of each sample was drawn into a normal plastic tube and centrifuged at $1000 \times g$ for 10 min. The serum obtained was stored at -20°C until analysis for biochemical metabolites and cytokines.

Analysis of serum biochemical metabolites and cytokines

Serum triglyceride, total cholesterol and low-density lipoprotein were analysed using a CX-4 automatic blood biochemical analyser (Beckman Inc., Fullerton, CA, USA) according to the reagent kit manufacturer's (Beijing Leadman Biochemistry Technology Co. Ltd, Beijing, China) instructions. Serum IL-2 and tumour necrosis factor (TNF) were determined using radioimmunoassay kits (Tianjing Jiuding Medical Biotech Co., Ltd, Tianjing, China) as described by Kong *et al.*^{11,13}

Analysis of lymphocyte proliferation activity

Peripheral blood lymphocytes were prepared as described by Kong *et al.*^{8,11,13} The cells were diluted to $2.5 \times 10^6 \text{ mL}^{-1}$ with RPMI 1640 medium (Gibco, NY, USA) supplemented with 100 IU L^{-1} benzylpenicillin, 100 IU L^{-1} streptomycin and 100 mL^{-1} foetal bovine serum. For analysis of proliferation activity, 90 μL of cell suspension and 10 μL of 80 $\mu\text{g mL}^{-1}$ concanavalin (a specific T cell mitogen) were added to a well in a 96-well tissue culture plate. All assays were performed in quadruplicate. After a 44 h incubation at 37°C in a 50 mL L^{-1} CO_2 incubator, 20 μL of 5 mg mL^{-1} 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) (Amresco Inc., Solon, OH, USA) was added to each well. This was followed by a 4 h incubation and then addition of 100 μL of dimethyl sulfoxide (Zheng-xing Institute of Chemical Engineering, Suzhou, China). The solutions were mixed for 10 min to completely dissolve the precipitates. Light absorbance was measured at 570 nm using an MQX200 enzyme-linked immunosorbent assay (ELISA) reader (Bio-Tek Instruments, Inc., Solon, OH, USA).

Statistical analysis

Data were expressed as mean $\pm$ standard deviation. Results were statistically analysed using one-way analysis of variance (ANOVA) (SAS Institute, Cary, NC, USA). Duncan's multiple range test was used to compare differences among treatment groups. A P value of less than 0.05 was taken to indicate statistical significance.

RESULTS

Growth performance in weanling rats after supplementation

The effects of CYP 3 on the average daily feed intake, average daily gain and feed/gain ratio in weanling rats are shown in Table 2. The results indicated that 0.25 or 0.5 g kg^{-1} CYP 3 supplementation increased ($P < 0.05$) the average daily feed intake and average daily gain on day 7 compared with 1 g kg^{-1} CYP 3 supplementation or no supplementation. On day 21 the 0.25 g kg^{-1} CYP 3-supplemented group showed higher ($P < 0.05$) average daily feed intake and average daily gain than the other three groups, while its feed/gain ratio was significantly lower ($P < 0.05$).

Serum contents of biochemical metabolites in weanling rats after supplementation

The effects of CYP 3 on the serum contents of biochemical metabolites in weanling rats are shown in Table 3. The results

Table 2. Effects of CYP 3 on growth performance in weanling rats

CYP 3 (g kg^{-1})	Average daily gain (g day^{-1})	Average daily feed intake (g day^{-1})	Feed/gain ratio
<i>Day 7 after initiation of supplementation (n = 10)</i>			
0	$5.31 \pm 0.37\text{c}$	$22.77 \pm 1.29\text{c}$	4.30 ± 0.08
0.25	$6.27 \pm 0.88\text{a}$	$25.34 \pm 0.76\text{a}$	4.11 ± 0.54
0.5	$5.98 \pm 0.69\text{ab}$	$24.01 \pm 1.22\text{b}$	4.05 ± 0.29
1	$5.45 \pm 0.73\text{bc}$	$22.32 \pm 1.48\text{c}$	4.13 ± 0.30
<i>Day 21 after initiation of supplementation (n = 6)</i>			
0	$6.69 \pm 0.47\text{b}$	$27.32 \pm 1.34\text{ab}$	$4.09 \pm 0.15\text{a}$
0.25	$8.02 \pm 0.49\text{a}$	$29.12 \pm 0.97\text{a}$	$3.64 \pm 0.19\text{b}$
0.5	$7.03 \pm 0.37\text{b}$	$27.32 \pm 1.48\text{ab}$	$3.89 \pm 0.10\text{a}$
1	$6.49 \pm 0.55\text{b}$	$25.88 \pm 1.94\text{b}$	$3.99 \pm 0.22\text{a}$

Values are mean $\pm$ standard deviation. For each day, means with different letters within a column differ significantly ($P < 0.05$).

Table 3. Effects of CYP 3 on serum contents of biochemical metabolites (mmol L^{-1}) in weanling rats

CYP 3 (g kg^{-1})	Low density lipoprotein	Total cholesterol	Triglyceride
<i>Day 7 after initiation of supplementation (n = 4)</i>			
0	$0.77 \pm 0.21\text{a}$	$2.48 \pm 0.27\text{a}$	$1.78 \pm 0.12\text{a}$
0.25	$0.54 \pm 0.08\text{b}$	$1.92 \pm 0.26\text{b}$	$1.58 \pm 0.02\text{b}$
0.5	$0.56 \pm 0.07\text{b}$	$2.04 \pm 0.06\text{b}$	$1.51 \pm 0.04\text{b}$
1	$0.51 \pm 0.08\text{b}$	$1.80 \pm 0.31\text{b}$	$1.50 \pm 0.10\text{b}$
<i>Day 21 after initiation of supplementation (n = 4)</i>			
0	$0.52 \pm 0.12\text{a}$	$2.24 \pm 0.45\text{a}$	$1.58 \pm 0.14\text{a}$
0.25	$0.43 \pm 0.06\text{ab}$	$1.87 \pm 0.25\text{ab}$	$1.45 \pm 0.08\text{ab}$
0.5	$0.40 \pm 0.06\text{ab}$	$1.55 \pm 0.33\text{b}$	$1.41 \pm 0.06\text{b}$
1	$0.35 \pm 0.05\text{b}$	$1.68 \pm 0.33\text{b}$	$1.43 \pm 0.10\text{ab}$

Values are mean $\pm$ standard deviation. For each day, means with different letters within a column differ significantly ($P < 0.05$).

indicated that CYP 3 supplementation at all three levels decreased ($P < 0.05$) the serum contents of low-density lipoprotein, total cholesterol and triglyceride on day 7 compared with no supplementation, and there were no significant differences ($P > 0.05$) between the three treatment groups. On day 21 the serum content of low-density lipoprotein was lower ($P < 0.05$) in the 1 g kg^{-1} CYP 3-supplemented group than in the other groups, while the serum contents of total cholesterol and triglyceride in the 0.5 and 1 g kg^{-1} CYP 3-supplemented groups were lower ($P < 0.05$) than in the other two groups.

Serum contents of cytokines in weanling rats after supplementation

The effects of CYP 3 on the serum contents of cytokines in weanling rats are shown in Figs 1 and 2. The results indicated that 0.5 and 1 g kg^{-1} CYP 3 supplementation increased ($P < 0.05$) the serum content of IL-2 on days 7 and 21 as well as the serum content of TNF on day 21 compared with 0.25 g kg^{-1} CYP 3 supplementation or no supplementation. The serum content of TNF on day 7 also increased ($P < 0.05$) in the 1 g kg^{-1} CYP 3-supplemented group compared with the non-supplemented group.

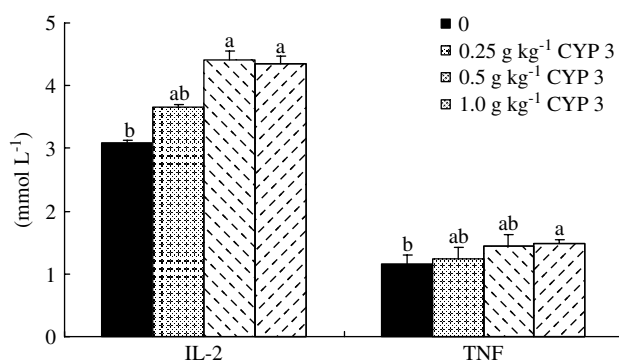


Figure 1. Effects of CYP 3 on serum contents of cytokines on day 7 after initiation of supplementation in weanling rats ($n = 4$). Values are mean $\pm$ standard deviation. For each cytokine, means with different letters differ significantly ($P < 0.05$).

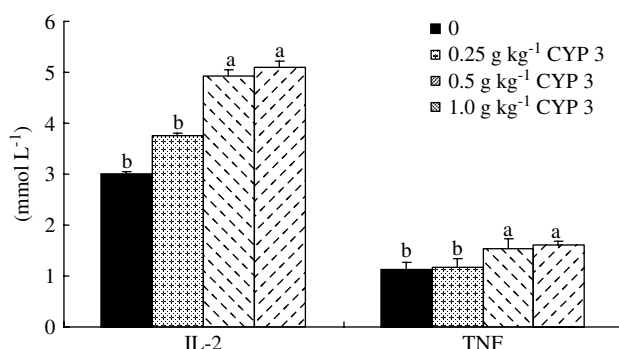


Figure 2. Effects of CYP 3 on serum contents of cytokines on day 21 after initiation of supplementation in weanling rats ($n = 4$). Values are mean $\pm$ standard deviation. For each cytokine, means with different letters differ significantly ($P < 0.05$).

Proliferation activity of peripheral lymphocytes in weanling rats after supplementation

The effects of CYP 3 on the proliferation activity of peripheral blood lymphocytes in weanling rats are shown in Fig. 3. The results indicated that 0.25 g kg⁻¹ CYP 3 supplementation increased ($P < 0.05$) the proliferation activity of lymphocytes on days 7 and 21 compared with the other three groups, 0.5 g kg⁻¹ CYP 3 supplementation increased ($P < 0.05$) the proliferation activity of lymphocytes on day 7 compared with the 1 g kg⁻¹ CYP

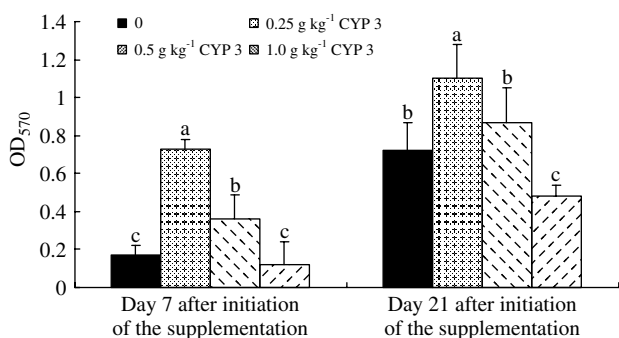


Figure 3. Effects of CYP 3 on lymphocyte proliferation activity in weanling rats ($n = 4$). Values are mean $\pm$ standard deviation. For each day, means with different letters differ significantly ($P < 0.05$).

3-supplemented and non-supplemented groups, while 1 g kg⁻¹ CYP 3 supplementation decreased ($P < 0.05$) the proliferation activity of lymphocytes on day 21 compared with the non-supplemented group. The stimulative effects in all treatment groups were higher ($P < 0.05$) on day 21 than on day 7.

DISCUSSION

There has been growing interest in recent years in the use of Chinese herbal medicines or their ingredients as new, alternative growth promoters for domestic animals and poultry.^{8,9,11,21–24} The present study is the first to determine the serum contents of biochemical metabolites (triglyceride, total cholesterol and low-density lipoprotein) and cytokines (IL-2 and TNF), the proliferation activity of peripheral blood lymphocytes as well as the growth performance in weanling rats supplemented with CYP 3. The results showed that dietary supplementation with 0.25 g kg⁻¹ CYP 3 increased the average daily feed intake and average daily gain but decreased the feed/gain ratio overall compared with the non-supplemented group. CYP 3 also significantly increased the serum contents of IL-2 and TNF and the proliferation activity of peripheral lymphocytes but decreased the serum contents of low-density lipoprotein, total cholesterol and triglyceride compared with the non-supplemented group.

The main function of the gastrointestinal system is to assimilate nutrients from the external environment into the animal's internal environment, where they are used for tissue growth and repair and for energy production.²⁵ Growth of animals is an outcome of complex metabolic transformations, including glucose and amino acid utilisation, intracellular protein turnover and fat deposition as well as their regulation by hormones and other factors.²⁶ Consistent with this view, dietary supplementation with 0.25 and 0.5 g kg⁻¹ CYP 3 enhanced the average daily feed intake and average daily gain on day 7 by 11 and 18% and by 5 and 13% respectively compared with the non-supplemented group, while dietary supplementation with 0.25 g kg⁻¹ CYP 3 enhanced the average daily feed intake and average daily gain on day 21 by 7–13% and by 14–24% respectively but reduced the feed/gain ratio by 6–11% compared with the other three groups. In addition, the dose of 0.25 g kg⁻¹ CYP 3 presented the strongest enhancing effects on growth performance in weanling rats. This herbal polysaccharide may improve the digestion and absorption of dietary nutrients and may also directly regulate the metabolism of absorbed nutrients through signal transduction mechanisms. Future studies are required to test this hypothesis.

Hyperlipidaemia is a major cause of mortality throughout the world.²⁷ From epidemiological and experimental investigations it is well known that diet plays an important role in the regulation of cholesterol homeostasis. Modern research shows that *R. dioscoreae* has the ability to lower blood lipids. Chen *et al.*²⁸ reported that a 250 or 500 g kg⁻¹ yam diet decreased the serum total cholesterol, triglyceride and low-density lipoprotein cholesterol levels in normal mice. An aqueous decoction of Chinese yam also decreased the serum total cholesterol and triglyceride levels in hyperlipidaemic mice.²⁹ A salient observation from the present study is that dietary supplementation with CYP 3 affected the serum contents of some metabolites (Table 3). Notably, CYP 3 decreased the serum contents of low-density lipoprotein, total cholesterol and triglyceride on day 7 after initiation of supplementation compared with the non-supplemented group. The serum content of low-density lipoprotein in rats fed 1 g kg⁻¹ CYP 3 on day 21 after initiation of supplementation was lower

in comparison with the other three groups, as were the serum contents of total cholesterol and triglyceride in the 0.5 and 1 g kg⁻¹ CYP 3-supplemented groups in comparison with the other two groups. Consistent with our findings, Wang *et al.*¹⁸ reported that Chinese yam starch significantly decreased the serum total cholesterol, triglyceride and low-density lipoprotein cholesterol levels in hyperlipidaemic rats. It is possible that this Chinese herb increases fatty acid oxidation in the skeletal muscle and liver by modulating the expression of key genes for lipid-metabolic enzymes.²⁶

The immune system, especially acquired immunity, plays an important role in protecting young animals against pathogenic infection. However, acquired immunity is underdeveloped at the age of weaning.³⁰ Thus exposure of weanling animals to various pathogens results in reduced productivity.^{31,32} A severe problem is the diagnosis and management of subclinical infections, especially when the pathogens are unknown. Therefore it is imperative that appropriate measures be taken to enhance the host's ability to resist a wide variety of pathogens.³³ The innate immune system is potentially useful in this regard, as it is non-specific and recognises a large number of different pathogens with a restricted set of receptors.³⁴ In this regard, components of innate immunity may serve as predictors of overall immunity as well as animal health. For this reason, we determined the proliferation activity of lymphocytes and the serum contents of cytokines in CYP 3-supplemented and non-supplemented weanling rats.

Lymphocytes are important circulating immune cells that play a critical role in maintaining immune functions and regulating inflammation.³⁵ Lymphocyte proliferation, a critical event of immune response in the host, was markedly increased in rats in response to dietary supplementation with CYP 3 in the present study. In particular, 0.25 g kg⁻¹ CYP 3 supplementation increased the proliferation activity of lymphocytes on days 7 and 21 compared with the other three groups, while 0.5 g kg⁻¹ CYP 3 supplementation increased the proliferation activity of lymphocytes on day 7 compared with the 1 g kg⁻¹ CYP 3-supplemented and non-supplemented groups. The stimulative effects in all treatment groups were higher on day 21 than on day 7 after initiation of supplementation. In addition, 1 g kg⁻¹ CYP 3 decreased the proliferation activity of lymphocytes on day 21 compared with the non-supplemented group, which may be related to other ingredients contained in CYP 3 (the carbohydrate content in CYP 3 was 909.1 g kg⁻¹). Obviously, the stimulative effects of CYP 3 on lymphocyte proliferation activity decreased with increasing dose, the strongest effect occurring at a dose of 0.25 g kg⁻¹, which clearly demonstrates that an appropriate dose is critical for eliciting a beneficial effect on lymphocyte proliferation and function.

The immune system is regulated by a highly interactive network of chemical communications, which includes the synthesis of antigen-presenting machinery, immunoglobulins and cytokines.³⁶ Cytokines play a central role in the cell-mediated immune response and also participate in the maintenance of tissue integrity. IL-2, as a kind of T cell growth factor, can induce the differentiation and stimulate the reactivity of many kinds of cells, e.g. cytotoxic T lymphocytes and lymphokine activated killers, and promote B cells to proliferate, differentiate and secrete immunoglobulins. In addition, IL-2 can stimulate the expression of the major histocompatibility complex (MHC) II type antigen, transferrin receptor and insulin receptor and generate manifold cytokines, including interferon (IFN)- γ , IL, TNF and so on. IFN- γ is another important cytokine generated mainly by CD4⁺ and CD8⁺ T cells;

it has antiviral effects and can regulate the immune response.³⁷ In this study, 0.5 and 1 g kg⁻¹ CYP 3 increased the serum contents of IL-2 on days 7 and 21 after initiation of supplementation, as well as the serum content of TNF on day 21, compared with the other groups. The serum content of TNF in the 1 g kg⁻¹ CYP 3 group also increased compared with the non-supplemented group. These findings collectively suggest that dietary supplementation with CYP 3 significantly enhances the immune response by increasing cytokine levels in weanling rats.

In summary, early weaning is a major stress factor that impairs intestinal metabolism and function and is associated with compromised immunity. Dietary supplementation with low-cost immunomodulatory phytochemicals may enhance the immune response in weanling rats. The results of the present study indicated an increase in cellular immunity by modulation of the production of cytokines in CYP 3-supplemented weanling rats, conferring an important protective role in the non-specific defence against infections. This herbal polysaccharide, as a natural green dietary additive for promoting the immune response and healthy growth, may offer an effective alternative to antibiotics for weanling animals.

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